PROJECT SYNOPSIS

A 3rd-Year Undergraduate Project Proposal

1. Project Title

Secure Reversible Data Hiding in Medical Images Using Block-Wise Histogram Shifting

2. Base Research Literature

Base Paper Title	Secure Reversible Data Hiding Using Block-Wise Histogram Shifting
Authors	Samar Kamil Khudhair, Monalisa Sahu, Raghunandan K. R., Aditya Kumar Sahu
Journal & Year	MDPI Electronics, Vol. 12, Issue 5, Article 1222 (March 2023)
Foundational Work	Reversible Data Hiding by Z. Ni et al. (IEEE T-CSVT, 2006)

3. Introduction & Context

Modern telemedicine platforms frequently transmit digital medical images (X-rays, MRIs, CT scans) across public networks and cloud servers. Alongside these scans, confidential patient records (identity, diagnostic notes, case history) must be securely sent to attending physicians.

Standard steganography methods permanently change pixel values. In medical diagnostics, even slight pixel modifications can be mistaken for lesions or obscure fractures, introducing legal and health risks. Reversible Data Hiding (RDH) solves this: it embeds confidential patient records inside the scan and, upon extraction, restores the scan back to its exact, untouched original state with zero permanent distortion (MSE = 0).

4. Problem Statement

- **Irreversible Image Degradation:** Standard methods permanently alter pixels, violating medical guidelines and diagnostic trust.
- **Capacity Constraints in Global Methods:** Traditional global histogram shifting treats the whole image as one unit, limiting embedding capacity.
- **Single-Layer Security Risk:** Hiding plaintext data without encryption leaves information vulnerable if the embedding is reversed.

5. Proposed Methodology

- **1. Block Partitioning:** The grayscale medical scan is divided into small, non-overlapping blocks (8×8 or 16×16) to leverage localized pixel correlation.
- **2. Histogram Shifting:** In each block, the algorithm identifies the peak bin and shifts intermediate pixels by ±1 to create space.
- **3. Payload Encryption & Embedding:** Patient notes are encrypted with a secret key and embedded into the peak pixels.
- **4. Lossless Restoration:** The recipient extracts the secret bits, decrypts the message, and reverses the pixel shifts (■1), restoring the original image bit-for-bit.

6. Key Project Objectives

- Develop a modular Python implementation of block-wise reversible data hiding.
- Guarantee mathematically proven lossless recovery (MSE = 0, PSNR = ∞ between original and restored images).
- Evaluate visual fidelity using standard benchmarks: PSNR (>52 dB) and SSIM (>0.99) for the stego image.
- Provide an interactive demonstration interface (Streamlit) for straightforward viva presentation.

7. Expected Deliverables

- Working Python program for block-wise embedding and extraction.
- Automated test script confirming 100% identical image restoration (np.array_equal == True).
- Demonstration dashboard displaying before/after histograms and evaluation metrics.